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Bonus Project
STAT 6340
Fall 2025
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Section 1: Report on questions

Question 1)

a) Please see the code part for the following architecture

i)One convolution layer with 32 3x3 filters (ReLu), followed by max-pooling over 2x2 blocks.

ii) One convolution layer with 64 3x3 filters (ReLu), followed by max-pooling over 2x2 blocks.

iii) One convolution layer with 64 3x3 filters. The shape of the output of this layer should be (3,3,64), then flattened (size 576)

iv) One dense layer of size(Relu)

v)Output dense layer with 10 units (softmax)

Please see the below output.

b) Parameter counts:

- conv2d_21 (Conv2D) →320 Parameters
- conv2d_22 (Conv2D) → 18,496
- conv2d_23 (Conv2D) → 36,928
- dense_12 (Dense) → 650
- Pooling and Flatten → 0
- Total Parameters = 93,322 parameters

Layer (type)	Output Shape	Param #
conv2d_21 (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d_18 (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_22 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_19 (MaxPooling2D)	(None, 5, 5, 64)	0
conv2d_23 (Conv2D)	(None, 3, 3, 64)	36,928
flatten_6 (Flatten)	(None, 576)	0
dense_12 (Dense)	(None, 64)	36,928
dense_13 (Dense)	(None, 10)	650

Total params: 93,322 (364.54 KB)

Trainable params: 93,322 (364.54 KB)

Non-trainable params: 0 (0.00 B)

c) The model achieved strong performance on the MNIST dataset, with a final test accuracy of 97.64%. We fitted the model in a) with 5 epochs and report its accuracy on the test data. Please see the below output. Validation accuracy is good on all of them. Validation accuracy is very close on Epochs2 to 5 (0.970 to 0.981), indicating minimal overfitting. Validation accuracy: achieved at97.64% and the Validation loss is between 0.084 to 0.149. Validation accuracy is high at epoch 5(0.9783)with lowest validation loss at epoch 1 with0.0843 and then rises

thereafter, suggesting mild overfitting.

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Epoch 1/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 46s 50ms/step - accuracy: 0.8184 - loss: 0.6129 - val_accuracy: 0.9745 - val_los
s: 0.0843
Epoch 2/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 79s 47ms/step - accuracy: 0.9750 - loss: 0.0869 - val_accuracy: 0.9683 - val_los
s: 0.1413
Epoch 3/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 41s 47ms/step - accuracy: 0.9783 - loss: 0.0820 - val_accuracy: 0.9682 - val_los
s: 0.1492
Epoch 4/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 39s 46ms/step - accuracy: 0.9801 - loss: 0.0768 - val_accuracy: 0.9768 - val_los
s: 0.1059
Epoch 5/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 47s 52ms/step - accuracy: 0.9811 - loss: 0.0783 - val_accuracy: 0.9783 - val_los
s: 0.1130
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - accuracy: 0.9701 - loss: 0.1217

Test accuracy: 97.640002%
```

Questions 2) **Note(I did problem 13 on chapter 10 from python textbook)**

13. Repeat the analysis of Lab 10.9.5 on the IMDb data using a similarly structured neural network. We used 16 hidden units at each of two hidden layers. Explore the effect of increasing this to 32 and 64 units per layer, with and without 30% dropout regularization.

To analyze how increasing the number of hidden units (from 16 to 32 and 64) and applying 30% dropout affects the performance of a two hidden layer neural network on IMDB sentiment classification. All models are used a two-hidden layer architecture, trained with a mini-batch size of 512 and a learning rate of 0.01.

Hidden Units	Dropout	Test Accuracy	Test Loss
16	0.0	0.8593	1.7610
32	0.0	0.8531	0.5276
64	0.0	0.8558	1.7286
16	0.3	0.8660	1.0663
32	0.3	0.8694	1.0839
64	0.3	0.8677	0.9234

Please refer the table above, we observe that without 30% dropout, increasing the number of hidden units did not consistently improve test accuracy. In fact, the highest test accuracy without dropout was achieved with 16 units. With 30% dropout, highest test accuracy achieved with 32 units (0.8694). On the other hand, the most significant drop in test loss occurred when increasing hidden units from 16 to 32 without dropout, substantially decreased 1.7610 to 0.5276.

We can conclude based on the results, increasing the units to 32 with 0 dropout, improved the test loss and increasing the units to 32 and dropout of 30% improved the test accuracy.